

1Q)

Actual class/ Predicted class $\rightarrow$	wolf (Warning) (yes)	No wolf/ Warning (No)	
wolf (yes)	Real wolf came, TP	wolf came, But Warning is not came FN	1
No wolf (No)	No wolf came, FP	No wolf TN No Warning	2
	3	0	3

- TP = 1 $\rightarrow$ Real wolf came, warning came
- FN = 0 $\rightarrow$ Real wolf came, but no warning
- FP = 2 $\rightarrow$ No wolf came, warning came twice
- TN = 0 $\rightarrow$ No wolf came, no warning

$$\therefore TP = 1$$

$$\therefore FP = 2$$

$$\therefore FN = 0$$

$$\therefore TN = 0$$

$$TP + FP + FN + TN$$

$$= 1 + 2 + 0 + 0$$

$$= 3$$

$$\therefore \text{Total} = 3$$

Q22

Actual / predicted	cat (Yes)	cat (No)	Total
cat (Yes)	TP 25	FN 5 20	45
No (cat)	FP 20 5	TN 50	55
Total	30	70	100

* actual image : 100 , TP, FP, FN, TN

- $TP = 25 \Rightarrow$ cat correctly predicted as cat
- $FN = 5 \Rightarrow$ cat incorrectly predicted as no cat
($30 - 25 \Rightarrow 5$)
- $TN = 50 \Rightarrow$ no cat correctly predicted as no cat
- $FP = 20 \Rightarrow$ no cat incorrectly predicted as cat ($70 - 50 = 20$)

Precision :

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{25}{25 + 20} \Rightarrow \frac{25}{45} = 0.555$$

$\Rightarrow 55.5\%$

Recall :

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{25}{25 + 20} \Rightarrow \frac{25}{45} = 0.555$$

$\Rightarrow 55.5\%$

F₁ Score :-

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.555 \times 0.555}{0.555 + 0.555}$$

$$= \frac{0.6185}{1.11} = 0.557 \Rightarrow 55.7\%$$

Data set

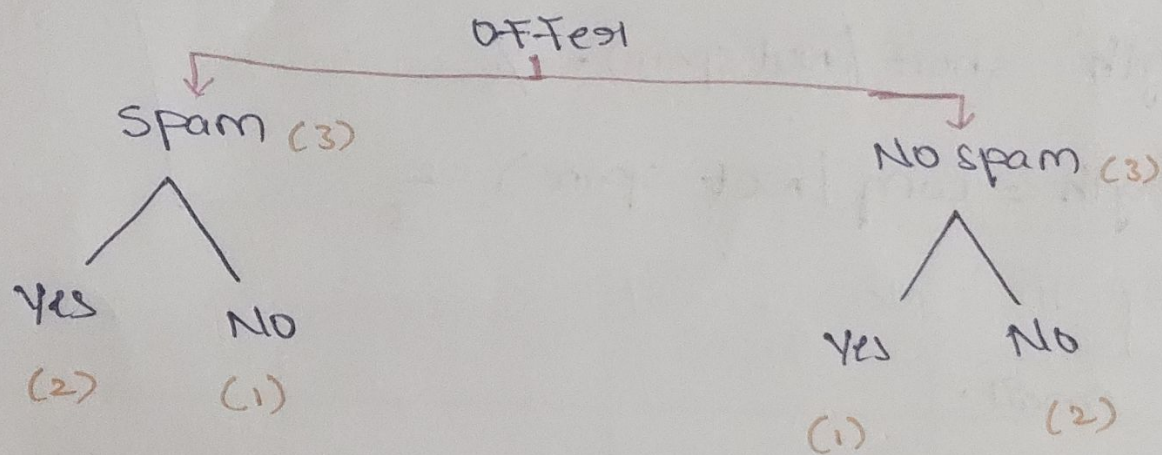
Offer	Free	Length	class
Yes	Yes	short	spam
Yes	No	long	Not spam
No	Yes	short	spam
No	No	long	Not spam
Yes	Yes	long	spam
No	No	short	Not spam

i) Prior probabilities.

Spam $\Rightarrow 3$; No spam $= 3$, Total $= 6$

$$P(\text{spam}) = \frac{3}{6} = 0.5 \quad , \quad P(\text{no spam}) = \frac{3}{6} = 0.5$$

ii) condition probabilities:

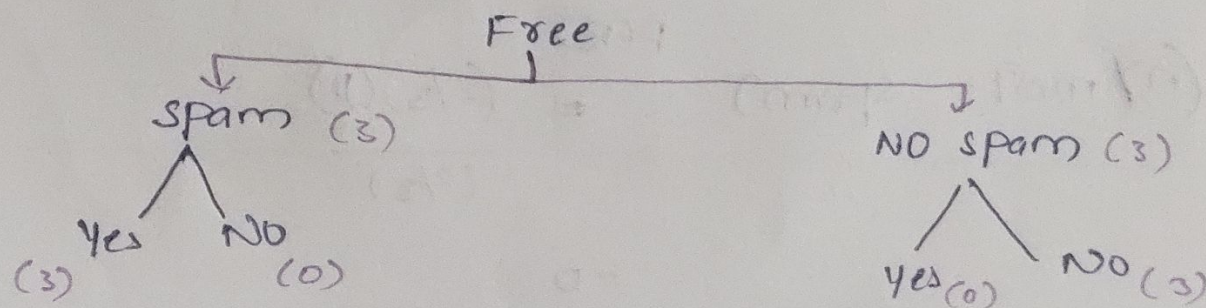


$$P(\text{Offer} = \text{yes} | \text{spam}) = \frac{2}{3}$$

$$; P(\text{Offer} = \text{yes} | \text{NO spam}) = \frac{1}{3}$$

$$P(\text{Offer} = \text{NO} | \text{spam}) = \frac{1}{3}$$

$$; P(\text{Offer} = \text{NO} | \text{NO spam}) = \frac{2}{3}$$

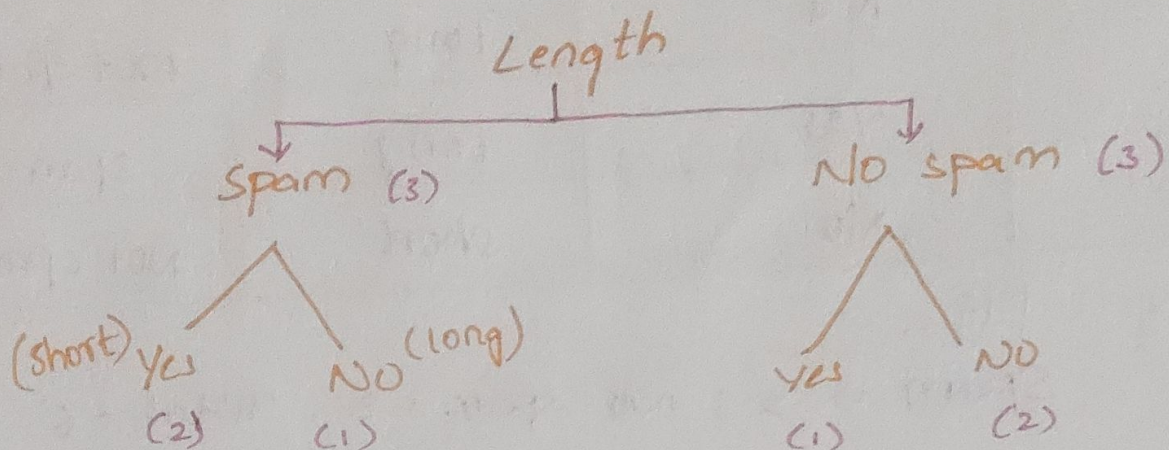


$$P(\text{Free} = \text{Yes} | \text{spam}) = \frac{3}{3} \Rightarrow 1$$

$$P(\text{Free} = \text{NO} | \text{spam}) = \frac{0}{3} \Rightarrow 0$$

$$P(\text{Free} = \text{Yes} | \text{not spam}) = \frac{0}{3} \Rightarrow 0$$

$$P(\text{Free} = \text{NO} | \text{not spam}) = \frac{3}{3} \Rightarrow 1$$



$$P(\text{length} = \text{short} | \text{spam}) = \frac{2}{3}$$

$$P(\text{length} = \text{long} | \text{spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{short} | \text{not spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{long} | \text{not spam}) = \frac{2}{3}$$

iii)

classify the New email

New email:

$X = (\text{Offer} = \text{Yes}, \text{Free} = \text{NO}, \text{length} = \text{short})$

Using Naive's Bayes:

$$P(A/B) = \frac{P(B|A) P(A)}{P(B)}$$

$$P(A | \text{mail} = \text{spam}) = \frac{P(B) P(A)}{P(B)}$$

$$= \frac{(\frac{2}{3})(\frac{1}{3})}{(\frac{2}{3})}$$

$$= 0$$

$$P(A / \text{mail} = \text{not spam})$$

$$\Rightarrow \left(\frac{1}{3}\right) \times 1 \times \frac{1}{3} \Rightarrow \frac{1}{9}$$

$$= 0.111$$

$$P(A / \text{mail} = \text{spam}) \times \text{probability}(\text{spam})$$

$$= 0 \times \frac{1}{2}$$

$$= 0$$

$$\text{Probability}(A / \text{mail} = \text{NO spam}) = \frac{1}{9} \times \frac{1}{2} = \frac{1}{18}$$

$$0.0556$$

④ Compare results using Weka (Naïves Bayes & classifier)

The data set Weka Experiments was used to compare three Bayesian classifiers

1) Bayes Net

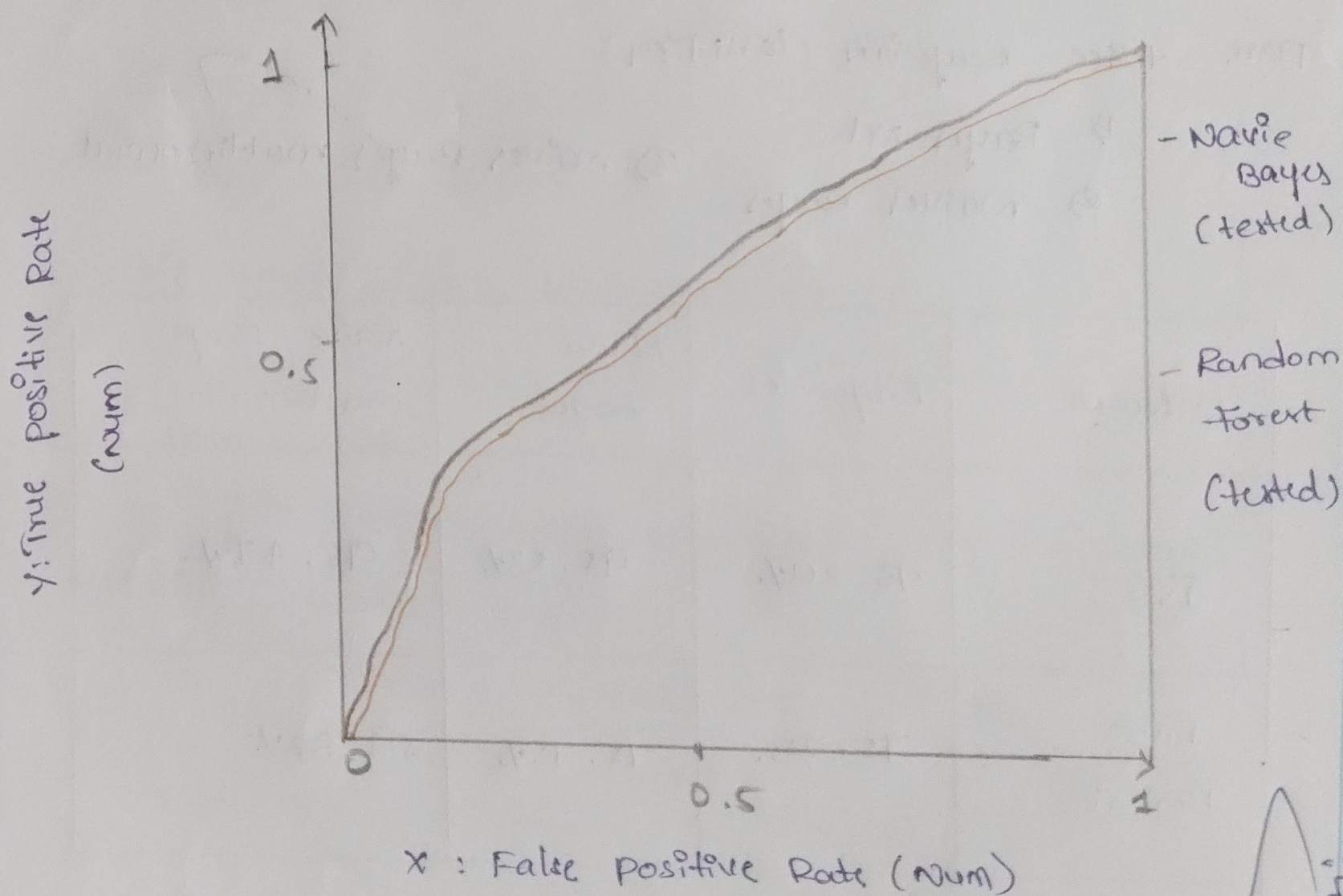
2) Naïves Bayes

3) Naïves Bayes multinomial

Dataset	Bayes Net	Naïve Bayes	Naïve Bayes multinomial
Iris	93.20%	95.53%	95.47%
Pima Diabates	75.25%	75.75%	59.82%

- For the Iris dataset, Naive Bayes gives 95.53% which is slightly higher than Bayes Net (93.20%) and Naive Bayes multinomial (95.47%).
- For the pima Diabetes dataset, Naive Bayes gives 75.75% which is higher than Bayes Net (75.25%) and Naive Bayes multinomial (59.82%).
- Therefore, Naive Bayes performs best among the three classifiers for both datasets shown in the WEKA output.

5



ROC curve - pima - diabetes

Shani